

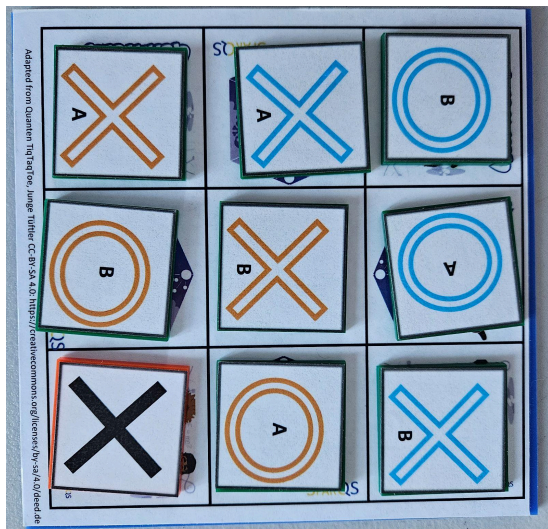
Quantum Tic Tac Toe Using Superposition:

The objective of this game is the same as classical Tic Tac Toe – get three in a row.

You have two options for making moves in this version of quantum tic tac toe.

1. Classical Moves → plays just like Classical Tic Tac Toe.
2. Superposition Move → place two X's or O's on the board at the same time (generally color coded to identify they are a pair with a label A and B to distinguish between the two) these two X's or O's are in a superposition. We won't know where our game piece is until we measure that superposition.

Play continues in turn order until the whole board is filled up with classical and quantum moves. As soon as the board is filled up the measuring process begins. The players take turns measuring all of the quantum moves (Measure each superposition one by one until they all become classical moves). It is important to note that each player can measure their own moves or their opponents moves. The measure process is done by rolling dice. Before you roll the dice to measure a superposition move you must identify which pair you are measuring. When all the quantum moves have been measured the game ends if someone won, but if no one has won the game the game continues by making moves to fill up the board once again, then measure when full and so on until someone wins the game or the game ends in a draw.

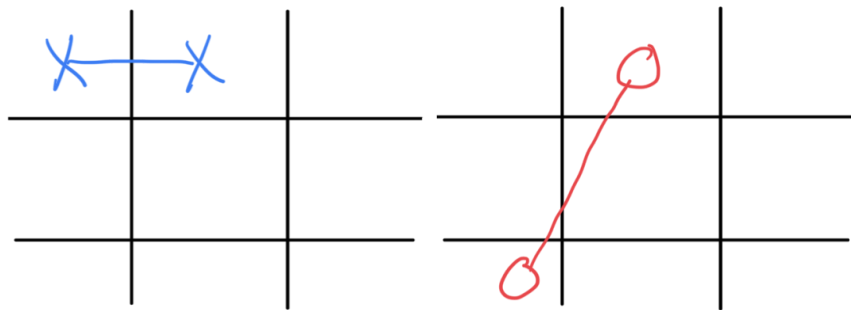


This is an example of a game board with all positions filled up. The Black X is an example of a classical move, while the other pieces are examples of superpositions (Two orange X's with labels A and B would be one example of a superposition move). It is important to note that X has not won even though they have three X's in a row in this image. They have not won because those X's are in superposition.

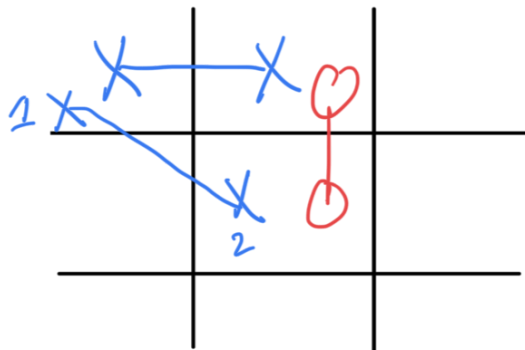
Quantum Tic Tac Toe using Entanglement:

The objective of this game is the same as classical Tic Tac Toe – get three in a row.

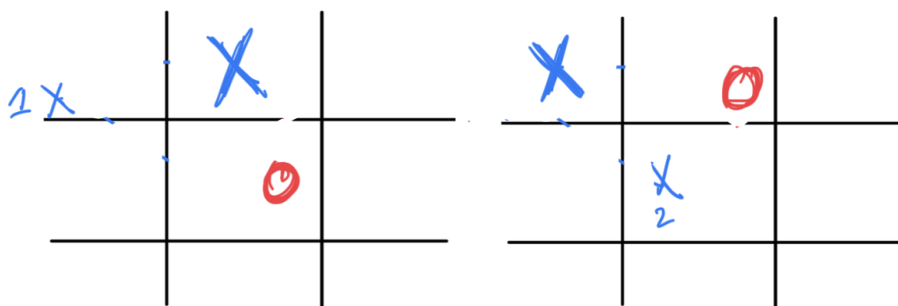
In this version there is only one option for your move. It is a quantum move where you entangle two X's or O's. You show this by a line between the two X's or O's. Below is an example of what this would look like. You can use any two positions on the board unless an entangled pair has been measured and has collapsed into a classical move (We will get the how we measure in a moment)



The measuring process happens as soon as a loop has been made. Below is an example of a loop. The player that closes a loop measures the last drawn pair of X's or O's with a dice or a coin. This will then have an impact with all other pairs in that loop (they will all collapse as well).

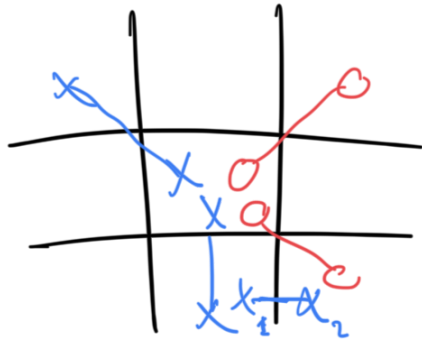


Notice in the above figure the labels of 1 and 2 are only on one pair. This is because that pair is the pair that closed the loop. Now we would roll a dice or flip a coin to measure that pair. In doing so it will collapse all other pairs in the loop. Below is an example of both scenarios for measuring 1 and 2.



Now that the measuring has happened those positions in the board have been locked, they are now classical (no entangled moves can be put in those positions any more). After the measuring happens the game will proceed in turn order, each player making more quantum moves until loops have been formed and measuring ensues until someone gets 3 in a row or the game ends in a draw.

Keep in mind some loops can get crazy and some loops can even affect some tangential moves. See below for an example of a loop affecting some other moves.



This bottom loop will collapse the two pairs that were not necessarily part of the loop.